

MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 11

1. *Let* $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ *and* $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$.

(1) Does $\mathbf{b}$ belong to the column space of A? Can you solve $A\mathbf{x} = \mathbf{b}$?

(2) What do you expect the projection of $\mathbf{b}$ onto $W = \text{Col}(A)$ to be?

(3) Find the projection $\hat{\mathbf{b}}$ of $\mathbf{b}$ onto $\text{Col}(A)$, and then solve $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$. (The vector $\hat{\mathbf{x}}$ is called *the least square solution* of $A\mathbf{x} = \mathbf{b}$.)

(4) Solve the equation $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$. Compare with your result of the previous part! (This equation is called the *normal equation* of $A\mathbf{x} = \mathbf{b}$.)

2. *True or false? If true, give a brief explanation why. If false, give a counterexample. All vectors below are in* $\mathbb{R}^n$ *for some* n.

(1) Let W be a subspace of $\mathbb{R}^n$ and $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in $\mathbb{R}^n$ such that $\mathbf{a} = \mathbf{b} + \mathbf{c}$ and $\mathbf{b}$ is in W and $\mathbf{c}$ is $W^\perp$. Then $\mathbf{b}$ is the projection of $\mathbf{a}$ onto W.

(2) All projection matrices are invertible.

(3) If $\mathbf{w}$ is a vector in W and T denotes the projection onto the subspace W, then $T(\mathbf{w}) = \mathbf{w}$.

(4) If $\mathbf{v}$ is a vector in $W^\perp$ and T denotes the projection onto the subspace W, then $T(\mathbf{v}) = 0$.

(5) If P be the projection matrix of a projection onto a subspace W, then $\text{Nul}(P) = W^\perp$.

(6) If $W = \mathbb{R}^n$, that is, we consider W to be the whole vector space $\mathbb{R}^n$, then the projection matrix (in the standard basis) is the identity matrix.

3. *Let* A *be a matrix with linearly independent columns. From class, we know that the projection matrix* P *of the projection onto the column space of* A *is* $A(A^TA)^{-1}A^T$.

(1) What is the projection matrix of the orthogonal projection onto the row space of A?

(2) What is the projection matrix for the orthogonal projection onto the left null space of A?

(3) Why is P symmetric (ie. $P^T = P$)?

4. *Let* $W = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$ *and* $V = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ *be subspaces of* $\mathbb{R}^3$.

(a) Find the projection matrices, P and Q, corresponding to the projections onto W and V, respectively.

(b) Check that $PQ = QP$. Can you interpret PQ as a projection matrix?

(c) Compute the matrices $P_{\mathcal{BB}}$ and $Q_{\mathcal{BB}}$ for projection onto W and B with respect to the input and output basis:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

(d) What is the product of $P_{\mathcal{BB}}$ and $Q_{\mathcal{BB}}$? How does your answer relate to the answer in part (b)?

5. *Recall that the orthogonal projection onto* $\mathrm{Col}(A)$ *has projection matrix* $A(A^TA)^{-1}A^T$.

(1) How does this formula simplify in the case when A has orthonormal columns?

(2) Let $Q = \begin{bmatrix} 1 & 0 \\ 0 & \frac{3}{5} \\ 0 & -\frac{4}{5} \end{bmatrix}$. What is the projection matrix corresponding to the orthogonal projection onto $\mathrm{Col}(Q)$?

(3) Let $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$. What is the projection matrix corresponding to the orthogonal projection onto $\mathrm{Col}(Q)$? Explain why your answer is not surprising.

6. *Let* A *be an* $m \times n$*-matrix*

(1) Explain why $\mathrm{Nul}(A) = \mathrm{Nul}(A^TA)$.

(2) Suppose A has linearly independent columns. Why is A^TA invertible?

7. (a) Given the following orthogonal set of vectors in $\mathbb{R}^4$:

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix} \right\}$$

use the FTLA (fundamental theorem of linear algebra) to complete the set to an orthogonal basis $\mathcal{B}$ for $\mathbb{R}^4$ (i.e., find a 4th vector to add to the set so that the set of 4 vectors is an orthogonal basis of $\mathbb{R}^4$). Hint: write the span, V, of this set as the column space of some matrix A, then find a basis for $V^\perp$.

(b) Let V be the span of the orthogonal set in part (a). Let $\pi_V : \mathbb{R}^4 \to \mathbb{R}^4$ be the projection onto V. Find the coordinate matrix $P_{\mathcal{B},\mathcal{B}}$ for π_V using $\mathcal{B}$ from part (a) as the input and output basis.

(c) Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be an orthogonal set of vectors in $\mathbb{R}^5$. Let $V = \mathrm{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and let $\mathcal{C} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ be an orthogonal basis for $\mathbb{R}^5$ (so $\mathcal{C}$ is the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ together with two additional vectors $\mathbf{v}_4$ and $\mathbf{v}_5$ which "completes it" to an orthogonal basis of $\mathbb{R}^5$). If $\pi_V : \mathbb{R}^5 \to \mathbb{R}^5$ is projection onto V, what is the matrix $P_{\mathcal{C},\mathcal{C}}$ for π_V with respect to input and output basis $\mathcal{C}$?

8. *A scientist tries to find the relation between the mysterious quantities* x *and* y*. She measures the following values:*

x	1	2	3	4
y	2	5	9	17

(i) Suppose that y is a linear function of x of the form $a + bx$. Set up the system of equations to find the coefficients a and b.

(ii) Find the best estimate for the coefficients a and b.

(iii) Same question if we suppose that y is a quadratic function of the $a + bx + cx^2$.

9. *Let A be an $m \times n$-matrix and let $\mathbf{b} \in \mathbb{R}^m$.*

(1) For which $\mathbf{b}$ is the following statement true:
If $\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2$ are least-squares solutions to $A\mathbf{x} = \mathbf{b}$, then $\hat{\mathbf{x}}_1 + \hat{\mathbf{x}}_2$ is a least-squares solution to $A\mathbf{x} = \mathbf{b}$.

(2) Let $\hat{\mathbf{x}}$ be a least-squares solution to $A\mathbf{x} = \mathbf{b}$ and $\mathbf{y}$ in $\mathrm{Nul}(A)$. Explain why $\hat{\mathbf{x}} + \mathbf{y}$ is a least-squares solution to $A\mathbf{x} = \mathbf{b}$.

10. *Quarterly economic data is subject to seasonal fluctuations. A curve that approximates the gross domestic product (GDP) of a country might be of the form*

$$y = \beta_0 + \beta_1 x + \beta_2 \sin(2\pi x/4),$$

where is x is the time in quarters of a year. The term $\beta_0 + \beta_1 x$ gives the basic GDP growth trend of the economy, while the sine term reflects the seasonal changes. Assume the GDP data are $(x_1, y_1), \ldots, (x_n, y_n)$.

(i) Give the design matrix that leads to a least-square fit to the equation above.

(ii) (Highly Optional) GDP data for US economy is available at http://www.bea.gov/. Using the above, can you find the GDP growth trend of the US economy.